

DRUGS AFFECTING FERTILITY AND REPRODUCTION

BIETE LUNDAU LUKE

DipPharm, BPharm, MclinPharm

OVERVIEW

2

- Human reproduction involves the cascade of hormonal secretion, the first being the gonadotropin releasing hormone (GnRH)
- GnRH stimulates the pituitary gland to secretion of two gonatropins namely , the follicle stimulating hormone (FSH) and the luteinizing hormone (LH)
- These gonadotropins further stimulate the production of steroids by gametes of the ovaries in the female and by the testis in the male
- Three categories of steroids produced are namely;
 - Estrogens e.g. estradiol, estrone, estriol
 - Progestins e.g. progesterone
 - Androgens e.g. testosterone

Overview Cont'd

- ▶ Estrogen, progesterone and testosterone are all produced in both female and males
- ▶ There is however, a marked differences in quantities produced in different sexes
- ▶ Females secrete more estrogens and progesterone while males produce more of testosterone

Physiological actions of estrogen and progesterone

4

- ▶ Estrogen and progesterone have multiple actions and interactions necessary for the reproduction
- ▶ Estrogen is produced in the granulosa cells of the ovary while progesterone is produced by the corpus luteum in response to LH secretion
- ▶ Estrogen promotes the growth and development of fallopian tubes, uterus and vagina as well as secondary sex characteristics like breasts, skeletal growth and axillary and pubic hair growth
- ▶ There are hormonal changes that occur during menstrual cycle where ovarian follicles are recruited in the follicular phase leading to the development of the dominant estrogen follicle
- ▶ In response to the estrogen positive feedback to the pituitary gland, a surge of LH is released at menstrual mid cycle which triggers ovulation
- ▶ In the luteal phase the follicle becomes the corpus luteum which secretes both estrogen and progesterone in response to LH

Physiologic function of estrogen and progesterone cont'd

5

- ▶ These two hormones prepare the uterus for implantation of fertilized ovum as the endometrium becomes more vascular and secretory
- ▶ If pregnancy does not occur, the corpus luteum ceases to produce estrogen and progesterone resulting into menstruation
- ▶ If pregnancy occurs, the placenta produces the human chorionic gonadotropin (hcG) which maintains the production of progesterone by the corpus luteum
- ▶ After about three(03) months, the placenta becomes the predominant source of progesterone, a hormone which is also referred to as the hormone of pregnancy as it prevents sloughing and miscarriage and thus maintain pregnancy
- ▶ Estrogen has a number of reproductive and bodily functions such as sensitization of the myometrium to oxytocin at parturition thereby, facilitating labor
- ▶ Furthermore, estrogen stimulates protein synthesis in the brain and hence affects mood and emotions, influences body fat distribution thus leading to development of female body contours, enhance blood coagulation by increasing the synthesis of clotting factors and prevent osteoporosis by inhibiting bone resorption

Physiologic function of estrogen and progesterone cont'd

- ▶ In males and females, estrogen is responsible for epiphyseal closure which halt linear bone growth
- ▶ While estrogen leads to the reduction of LDL and increase in HDL, progesterone causes a dose related increase in the LDL while reducing HDL
- ▶ Progesterone increase basal temperature by 0.5 to 0.8 celcius at ovulation and during the entire of luteal phase
- ▶ Progesterone also affects the emotional state and also presents with some mineralocorticoid effect (Salty retaining properties)

Physiologic actions of Testosterone

- ▶ LH stimulates testosterone synthesis in the leydig cells of the testes while FSH promotes spermatogenesis by sertoli cells in the seminiferous tubule
- ▶ Sertoli cells also produce a protein called **inhibin** which in collaboration with Dihydrotestosterone (DHT) act as feedback regulator of FSH by the pituitary
- ▶ Testosterone is responsible for the development of secondary sex characteristic in males at puberty which includes growth of the larynx, thickening of vocal cords and growth of facial, axillary and pubic hair
- ▶ Apart from growth stimulation of the penis, scrotum, seminal vesicles and prostate gland, testosterone stimulates and maintains sexual function in men (also hormone of men)
- ▶ Testosterone and other androgens also increase lean body mass, skeletal muscle growth, increased sebaceous gland activity and sebum production thus leading to development of acne in both sexes
- ▶ Increase erythropoietin growth in the kidneys and decrease HDL cholesterol

Estrogens preparations

- ▶ Naturally available estrogens are **estradiol** and **conjugated estrogen** (Sulphate ester of estron and equilin)
- ▶ **Micronized estradiol**
 - orally administered with good bioavailability
 - Also available as vaginal tabs or ring that slowly releases the drug
- ▶ **Ethinyl estradiol** and **mestranol** are a synthetic modification of estradiol by the addition of the ethyl group
 - Reduced first pass effect and increase the half life to about 20 hours
 - Evidence of improved oral potency than estradiol itself
- ▶ Long acting formulations of estradiol are also available meant for transdermal or intramuscular administration

Indications of estrogens

- ▶ Estradiol or mestranol are combined with progesterone and used as contraceptives, menstrual disorders like menorrhagia, dysmenorrhea and also control of acne
- ▶ Primary hypogonadism which may be surgical oophorectomy, menopause etc
- ▶ Micronised estradiol, conjugated equine estrogen and transdermal estrogen are used in hormonal replacement therapy (HRT) in post menopausal women

Adverse effects of estrogen

The following are the most common dose pendent adverse effects and not very common in the lower doses used in HRT;

- ▶ Breast tenderness
- ▶ Headache
- ▶ Edema
- ▶ Nausea
- ▶ Vomiting
- ▶ Anorexia
- ▶ Changes in libido

The following are the more serious adverse effects;

- ▶ Hypertension
- ▶ Thromboembolic disorders e.g. DVT
- ▶ Gall bladder disease

Contraindications/ Cautions in the use of estrogens

- ▶ Pregnancy
- ▶ Should be avoided in women with uterine fibroids
- ▶ Great caution to be taken in women with hepatic disease, endometriosis, thromboembolic diseases or hypercalcemia

Progesterones

- ▶ Progesterone is the primary natural progestin in mammals
- ▶ It usually undergoes an extensive first pass effect and has a short half life
- ▶ To improve on its pharmacokinetic properties, esters of progesterone have been produced
- ▶ Examples of these progesterone esters are **megestrol**, **hydroxyprogesterone caproate** and **medroxyprogesterone acetate**
- ▶ Megestrol is administered orally, hydroxyprogesterone is administered as a long acting intramuscular injection while medroxyprogesterone can be administered both orally and intramuscularly

Indications of progesterone

- ▶ Dysmenorrhea, endometriosis and uterine bleeding through the feedback inhibition of gonadotropin secretion by the pituitary gland
- ▶ HRT in combination with estrogen where they help reduce incidence of estrogen induced irregular bleeding and prevention of uterine hyperplasia and endometrial cancer
- ▶ Hydroxyprogesterone caproate once weekly injections are approved to reduce the risk of preterm delivery in women with a history of at least one spontaneous preterm birth

Hormone replacement therapy(HRT)

- ▶ Menopause refers to the cessation of menstruation which mostly occur between the age of 45-55 years
- ▶ Evidence of menopause occurring earlier has also been recorded
- ▶ Slightly before menopause (Pre menopausal), the supply of eggs in the ovaries decline and ovulation becomes irregular
- ▶ The ovarian follicles fail to develop and hence leading to insufficient estrogen making it difficult to suppress FSH and hence its rising(FSH > 40IU/L confirms woman having reached menopause)

Symptoms of pre or post menopausal syndrome (PMS)

- ▶ Hot flashes
- ▶ Flashes that consists of alternating chills and sweating accompanied by vomiting
- ▶ Nausea
- ▶ Dizziness
- ▶ Headache
- ▶ Tachycardia and palpitations
- ▶ Episodes of these symptoms often occur several times a day but night sweats are particularly common

Management of PMS

- ▶ Estrogen can be used alone for HRT in women who have had hysterectomy
- ▶ In women with a uterus, a low dose estrogen should be used in combination with progesterone to reduce the risk of endometrial cancer
- ▶ Estrogen can be administered in different dosage forms and routes thus allowing women to try alternative preparations and finally determine their most effective and tolerable

Other documented benefits of Estrogen

- ▶ Proved to relieve other menopausal symptoms like urogenital, vulva and vaginal atrophy
- ▶ Improvement of mood and reduce cognitive difficulties probably due to the improved sleep

NB:

HRT should not be used in women with a history of endometrial cancer neither should it be used in breast cancer survivors

CONTRACEPTIVES

- ▶ Contraceptives are drugs or devices used to prevent conception and thus allowing women to plan and or space their children

There are different types of contraception;

- ▶ Hormonal – oral, IM injectable, long acting intradermal implants
- ▶ Barrier methods – condoms, diaphragm
- ▶ Intrauterine devices
- ▶ Spermicides – chemicals acting locally in the reproductive tract where they kill the sperms

Hormonal contraceptives

18

1. Oral contraceptives

a). Combined oral contraceptive

- ▶ These contain a combination of estrogen and progesterone which may either be monophasic or triphasic contraceptives
- ▶ The monophasic contains the same amount of progesterone through out the administration cycle while in the triphasic type, progesterone amounts are increased after day 7 and 14 of the cycle
- ▶ The newer contraceptives sequentially steps down the estradiol while progesterone is stepped up
- ▶ Common combinations are those containing ethinyl estradiol + levonorgestrel and those containing ethinylestradiol + drospirenone

Mechanism of action

- ▶ Estrogen-progesterone contraceptives act primarily by feedback inhibition of GnRH secretion
- ▶ This leads to decreased gonadotropins and subsequently inhibit ovulation
- ▶ Estrogen component reduce the FSH secretion and the selection and maturation of the dominant follicle
- ▶ The progesterone component prevent the mid cycle LH surge required for ovulation

Indications

- ▶ Contraceptives
- ▶ Menstrual disorders like dysmenorrhea
- ▶ Acne vulgaris

Common adverse effects of oral contraceptives

Estrogen excess	Progesterone effect	Estrogen deficiency	Progesterone deficiency
Breast enlargement	Acne	Atrophic vaginitis	Dysmenorrhea
Dizziness	Fatigue	Continuous bleeding	Hypermenorrhea
Dysmenorrhea	Hirsutism	Early or mid cycle bleeding	Late cycle bleeding
Edema	Libido change	Hypomenorrhea	
Headache	Oily skin	Vasomotor symptoms	
Irritability	Weight gain (Cyclic)		
Nausea and vomiting			
Weight gain (Cyclic)			

Interactions

- ▶ Carbamazepine, phenytoin and other CYP 450 enzyme inducers increase the hepatic metabolism of oral contraceptives
- ▶ This may lead to reduced plasma levels of the contraceptive steroid which is likely to lead to contraceptive failure
- ▶ Antibiotics like penicillins and tetracycline can eradicate intestinal flora which is responsible for enterohepatic cycling of the contraceptive which may lead to them being ineffective
- ▶ Estrogen inhibits the metabolism and potentiate the effect of cyclosporine, antidepressants and glucocorticoids
- ▶ Estrogen increases the synthesis of vitamin K dependent clotting factors and so may antagonize the effect of warfarin

b). Progesterone only contraceptives

- ▶ Example include norethindrone and levonogestrel products that are administered orally and are sometimes called minipill

Mechanism of action of progesterone only contraceptives

- ▶ Thicken and decrease the amount of cervical mucus making it difficult for the sperm to swim and penetrate
- ▶ Also create a thin, atrophic endometrium which is hostile to implantation of the blastocyst

Beneficiaries of the progesterone only contraceptives

- ▶ Women who smoke
- ▶ Older women
- ▶ Women in whom estrogen is contraindicated

Comparison of Failure rates

- ▶ Higher failure rates have been reported with Progesterone only contraceptives compared to the estrogen-progesterone combined products

Emergency contraceptives

- ▶ These are also progesterone only containing, levonogestrel being the common drug
- ▶ Meant to prevent conception in women not on regular contraception methods or victims of abuse like rape cases
- ▶ These tablets have been called using different names i.e. morning after pill, 72hours etc
- ▶ The 0.75mg is to be take within the 72hours after sexual intercourse and the second dose to be taken 12hous after the first dose while the 1.5g tablet is sufficient as a single dose

Side effects

- ▶ Nausea which can be controlled by usage of anti-emetics
- ▶ Headache
- ▶ Dizziness
- ▶ Leg cramps
- ▶ Abdominal cramps

Non oral hormonal contraceptives

Injectables

- ▶ Norethisterone - injected every 2 months
- ▶ Medroxyprogesteron (Depoprovera) – injected 3 months respectively
- ▶ **Subdermal implants** like **jadelle** which slowly release progesterone and provide a coverage of up to 3 -5 years
- ▶ **Hormonal Intra uterine devices** like mirena, Skyla also slowly releases progesterone
- ▶ Depending on the type, these can provide a prevent pregnancy up to the period of 10 years

Transdermal patches (Estrogen-progesterone)

- ▶ Uses a 4 week administration cycle where with a new patch being applied weekly for 3 weeks
- ▶ Week 4 is patch free to allow withdrawal bleeding

END